

SIMATS ENGINEERING

ASSESSMENT TOOL 1 : ANALYTICAL PROBLEM SOLVING

CO2 : ASSESSMENT TOOL - I

Field	Details
Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA1516
Course Faculty	Dr. Santhiya Lakshmi Kumaravel
Student Name	Vino Aldrin E
Registration No.	192511481
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Contents

1. Comparative Analysis of Type-1 and Type-2 Hypervisors
2. Workload Isolation and Attack Surface Analysis
3. VM Host Utilization Snapshot and Corrective Actions
4. Software Defined Datacenter versus Traditional Datacenter
5. Multi-Cloud Federation Identity Design

Question 1 : Type-1 vs Type-2 Hypervisors for a Mission-Critical Cloud Datacenter

A hypervisor is the virtualization layer that allows multiple virtual machines to share the same physical hardware. Its placement relative to the hardware and the host operating system decides how it performs, how secure it is, and how it can be managed at scale.

Type-1 (Bare Metal) Hypervisor

Runs directly on the physical server hardware without an intervening host operating system. Examples include VMware ESXi, Microsoft Hyper-V and Citrix XenServer. Because the hypervisor talks to the hardware directly, resource scheduling for CPU, memory and I/O is handled by a thin, purpose-built kernel.

Type-2 (Hosted) Hypervisor

Runs as an application on top of a conventional host operating system, for example VMware Workstation, Oracle VirtualBox or Parallels Desktop. Every hardware request from a guest VM passes through the host OS scheduler before it reaches the hardware.

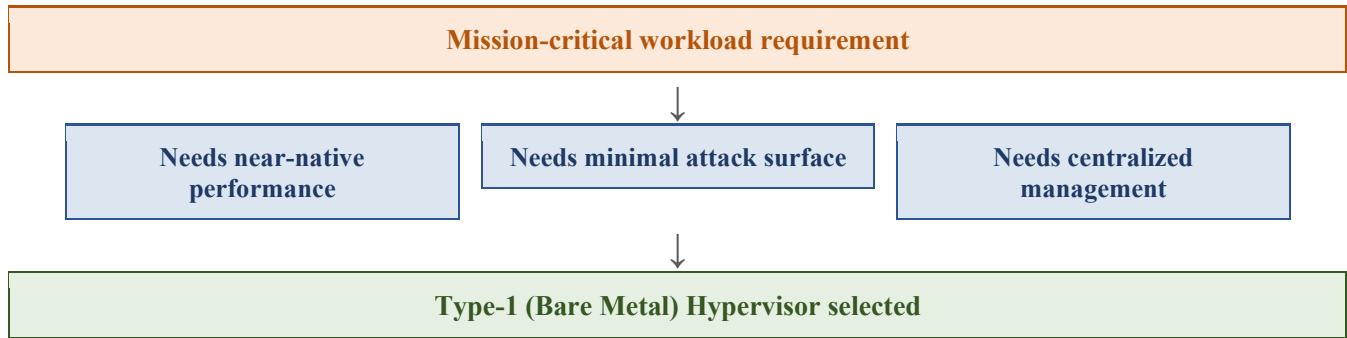
Comparative Analysis

Parameter	Type-1 Hypervisor	Type-2 Hypervisor
Performance	Near-native, direct hardware access, low scheduling overhead	Reduced, requests pass through host OS layer, higher latency
Security	Small attack surface, minimal codebase, strong VM isolation	Larger attack surface, inherits host OS vulnerabilities
Manageability	Centralized tools (vCenter, SCVMM), built for datacenter scale	Managed per machine, not suited for large fleets
Resource Efficiency	High consolidation ratio, efficient for many concurrent VMs	Lower, host OS consumes CPU and memory itself
Typical Use Case	Production cloud datacenters, mission-critical workloads	Development, testing, personal sandboxing

Table 1.1 Type-1 versus Type-2 hypervisor comparison

Justification

For a datacenter hosting mission-critical applications, the Type-1 hypervisor is the better choice. It removes the host operating system as a point of failure and as an attack surface, gives predictable low-latency performance under heavy consolidation, and integrates with enterprise management planes that support high availability, live migration and centralized patching. These properties directly support the uptime and compliance requirements that mission-critical workloads demand, which a hosted Type-2 hypervisor cannot guarantee at datacenter scale.

Figure 1.1 Selection Logic for a Mission-Critical Datacenter

Decision flow showing why a bare metal hypervisor satisfies all three requirements

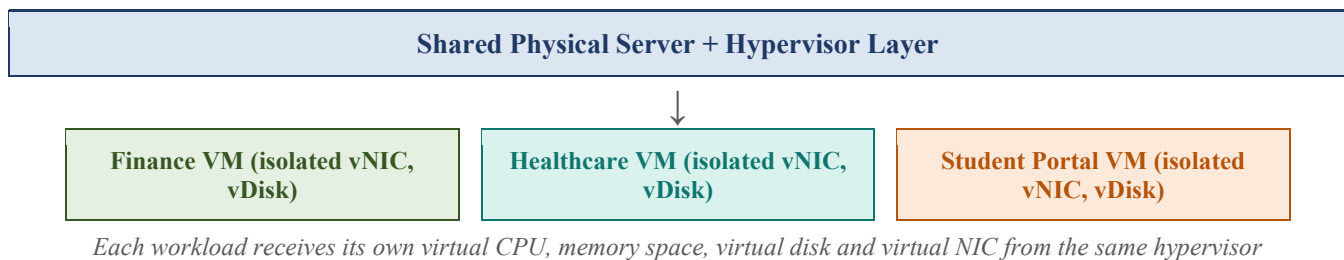
Question 2 : Isolating Finance, Healthcare and Student Portal Workloads

When a single cloud provider hosts finance, healthcare and student portal workloads on shared hardware, virtualization architecture is what keeps one tenant's data and processes separate from another's, even though they physically share CPU, memory, storage and network paths.

How Virtualization Supports Isolation

- Compute isolation: the hypervisor uses hardware-assisted virtualization (Intel VT-x, AMD-V) and extended page tables so each VM has its own memory address space that other VMs cannot read or write.
- Storage isolation: each tenant is assigned separate virtual disks or datastores, with LUN masking and access control lists preventing cross-tenant storage access.
- Network isolation: virtual switches, VLANs and security groups segment traffic so packets from the finance VM cannot reach the healthcare or student VM without passing through an explicit routing and firewall policy.
- Policy isolation: role-based access control and separate management contexts stop an administrator scoped to one tenant from touching another tenant's resources.

Figure 2.1 Workload Isolation on Shared Hardware



Two Major Attack Surfaces

Attack Surface	Description
Hypervisor / VM escape	A flaw in the hypervisor lets code running inside one VM break out and access the hypervisor or a co-resident VM, defeating the isolation the architecture is meant to provide. This is the most severe risk because a single hypervisor compromise exposes every tenant on that host.
Shared virtual network layer	A misconfigured virtual switch, VLAN, or security group can allow traffic to leak between tenants, and co-resident VMs can be used for side-channel attacks such as cache-timing analysis to infer another tenant's data.

Table 2.1 Primary attack surfaces introduced by multi-tenant virtualization

Question 3 : VM Host Utilization Snapshot Analysis

The snapshot reports CPU at 92 percent, memory at 88 percent, storage I/O latency at 35 ms, and an average VM boot time of 170 seconds. Read together, these four numbers point to a host that is resource-starved rather than affected by a single isolated fault.

Figure 3.1 Utilization Snapshot

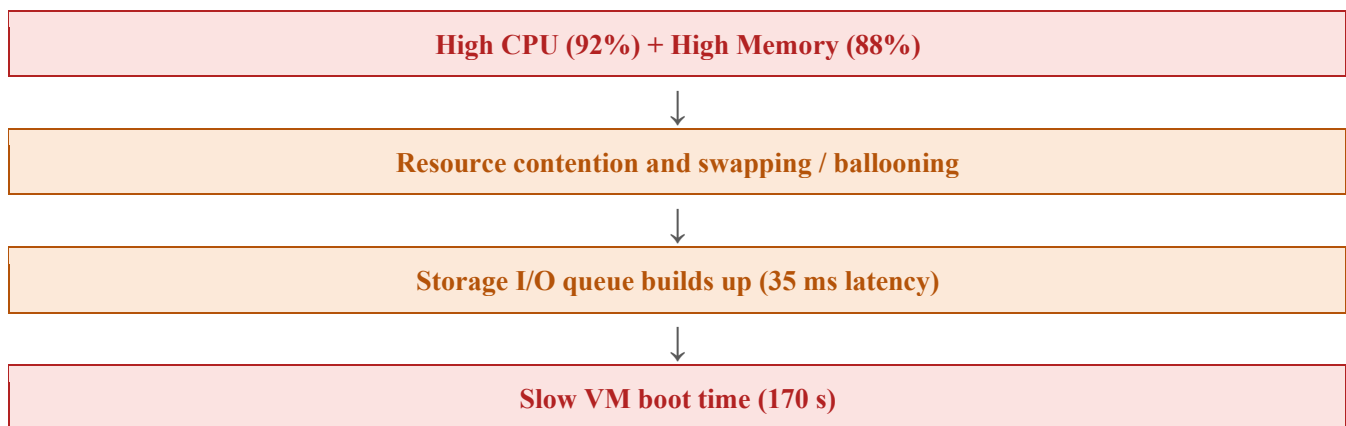


Storage I/O and boot time bars are scaled against acceptable enterprise thresholds (10 ms and 60 s respectively), not against 100

Likely Bottlenecks

- CPU at 92 percent is close to saturation. VMs begin to queue for CPU cycles, which shows up as scheduling contention, often called the noisy neighbor effect.
- Memory at 88 percent leaves little headroom. The hypervisor is likely relying on memory ballooning or swapping to the host disk to satisfy guest requests, both of which slow every VM on the host.
- Storage I/O latency of 35 ms is well above the 5 to 10 ms expected from enterprise SSD or NVMe backed storage, indicating the storage layer cannot service the current IOPS demand.
- The 170 second boot time is the visible symptom of the above three constraints acting together: a VM booting needs CPU cycles, memory allocation and heavy disk reads at the same time, so it is the first place contention becomes visible to users.

Figure 3.2 Root Cause Chain



Compute pressure and storage pressure compound each other during boot, when all three resources are requested at once

Recommended Corrective Actions

Resource	Corrective Action
CPU	Apply CPU reservations or limits for critical VMs, enable DRS style load balancing, or add a host to the cluster to spread the load.

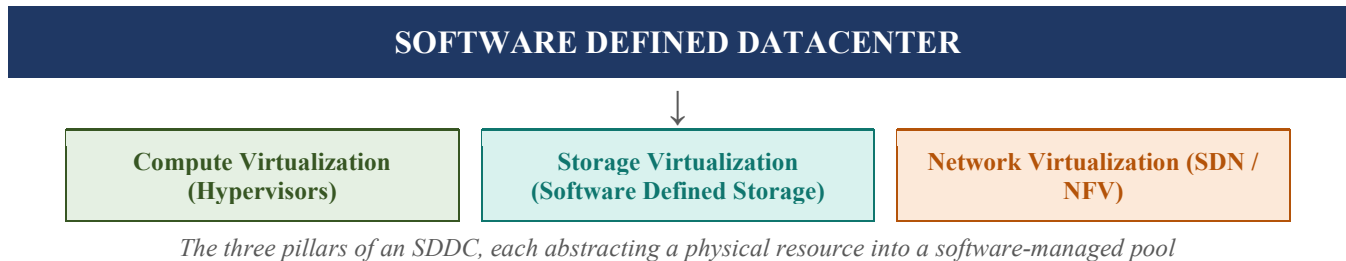
Resource	Corrective Action
Memory	Increase physical RAM on the host, right-size over-provisioned VMs, and enable transparent page sharing to reduce duplicate memory pages.
Storage	Move latency-sensitive VMs to a faster tier such as NVMe or all-flash storage, and enable storage I/O control to stop a single VM from monopolising IOPS.
Boot Time	Stagger scheduled reboots to avoid boot storms, and use paravirtualized drivers to reduce the CPU and I/O cost of the boot sequence.

Table 3.1 Corrective actions mapped to each observed bottleneck

Question 4 : Software Defined Datacenter versus Traditional Datacenter

A traditional datacenter provisions compute, storage and network as separate, hardware-bound silos. Each change, such as adding a server or opening a network path, requires manual configuration on physical equipment and can take days or weeks. A Software Defined Datacenter (SDDC) abstracts all three layers into software so they can be provisioned, scaled and reconfigured through policy and automation instead of manual hardware work.

Figure 4.1 SDDC Abstraction Model



Compute Virtualization

Hypervisors decouple the operating system and application from the underlying server, so a virtual machine or container can be provisioned in minutes instead of the days needed to rack and configure physical hardware. Workloads can also be live-migrated between hosts without downtime, which a traditional datacenter cannot do.

Storage Virtualization

Software Defined Storage pools capacity across many physical disks and nodes into a single logical resource, for example VMware vSAN or Ceph. Storage can then be allocated, tiered and replicated through policy, removing the need to manually provision LUNs on dedicated storage arrays for every new workload.

Network Virtualization

SDN and NFV replace physical cabling changes with virtual networks and overlays such as VXLAN. Firewall rules, load balancers and routing policies are deployed in software, which allows microsegmentation and rapid network provisioning that would otherwise need a network engineer to physically rewire switches.

Agility Gained

Capability	Traditional Datacenter	SDDC
Provisioning time	Days to weeks, manual hardware setup	Minutes, API or self-service driven
Scaling	Limited by physical hardware on site	Elastic, pools resources across the cluster
Configuration	Manual, device by device	Policy-driven, automated, repeatable
Recovery	Manual failover, hardware dependent	Automated failover and orchestrated DR

Table 4.1 Agility comparison between traditional and software defined datacenters

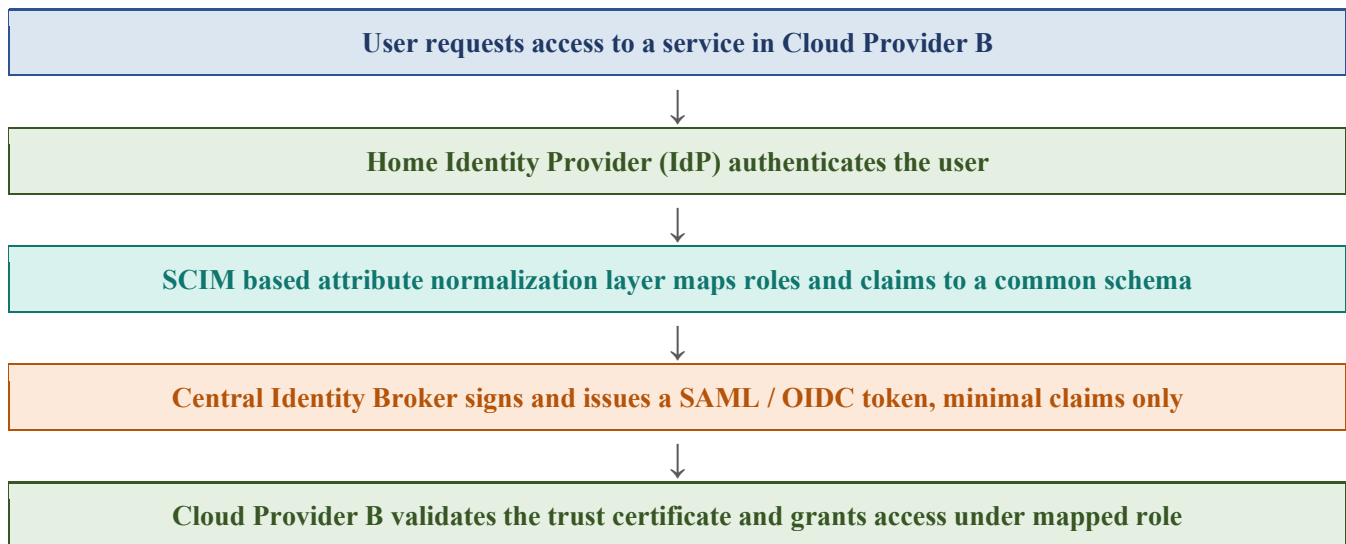
Question 5 : Multi-Cloud Federation Identity and Privacy Design

A multi-cloud federation project fails when a user authenticated with one provider is not correctly recognized by another. This usually happens because each provider maintains its own identity schema, its own attribute naming, and sometimes a different federation protocol, so an identity that is valid in Cloud A does not map cleanly onto Cloud B.

Analysis of the Failure

- Protocol mismatch: one provider may use SAML 2.0 while another expects OpenID Connect (OIDC), so authentication assertions are not understood on the receiving side.
- Attribute mismatch: role or group names differ between providers, for example "Finance-Admin" versus "finance_admin", so authorization decisions fail even after authentication succeeds.
- No central trust anchor: without a broker that both providers trust, each pair of providers would need a separate trust relationship, which does not scale and increases the chance of a missed or stale trust configuration.
- Privacy exposure: without attribute minimization, more user data than necessary is passed between providers, which increases the impact of a mismatch or a breach.

Figure 5.1 Proposed Federated Identity Architecture



A single trusted broker with attribute normalization removes the need for pairwise trust between every provider

Recommended Secure Identity and Privacy Design

Design Element	Purpose
Standard federation protocol (SAML 2.0 or OIDC)	Guarantees that authentication assertions are understood by every participating provider.
Central Identity Broker	Acts as the single trust anchor, so providers only need one trust relationship each instead of one per pair.
SCIM attribute normalization	Maps differing role and attribute names to one common schema before an assertion is issued.

Design Element	Purpose
X.509 signed assertions over TLS	Ensures tokens cannot be forged in transit and confirms the issuing party's identity.
Least privilege claim minimization	Only the attributes required for the specific request are shared, reducing privacy exposure.
Continuous audit logging	Cross-provider authentication events are logged centrally to detect anomalies such as repeated mismatches or replayed tokens.

Table 5.1 Design elements addressing both the identity mismatch and privacy requirements